CSC336 – Web Technologies | Lab Midterm Exam

Upgrade Your E-commerce Listing and Product Pages

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Parameters Derived from N	N = 63	Breakpoint (BP) = 663px	Categories (K) = 3
Accent Color (H)	hsl(81, 70%, 45%) [H = (63 × 7) mod 360 = 81°]
Offsets & Thresholds	Sticky Offset: (63 mod 40) + 70 = 93px	Premium Threshold: 63 × 100 = Rs. 6,300

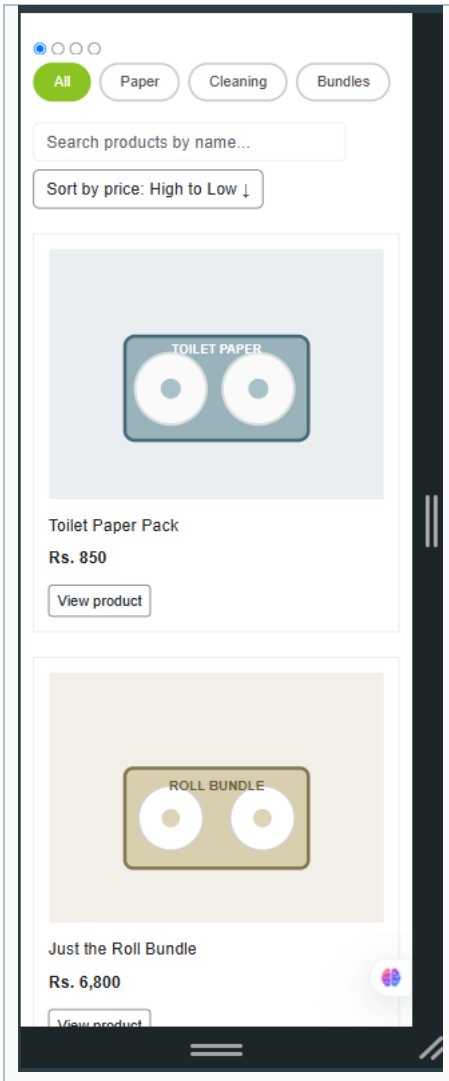
PART 1: SCREENSHOTS

Paste screenshots of each task working in the browser at 375px (mobile), 663px (BP), and 1280px (desktop) into the designated placeholder boxes below.

Task 1 — CSS-only Category Filter Screenshots

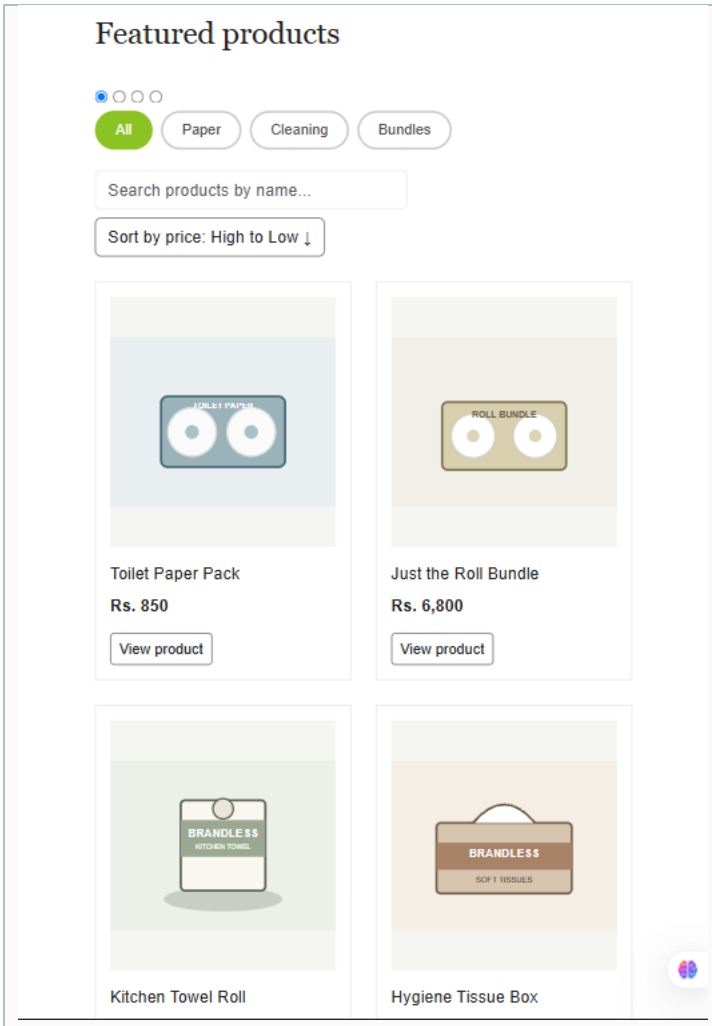
• Task 1 at 375px (Mobile View)

Category filter pills (All, Paper, Cleaning, Bundles) wrapping properly on small screens.



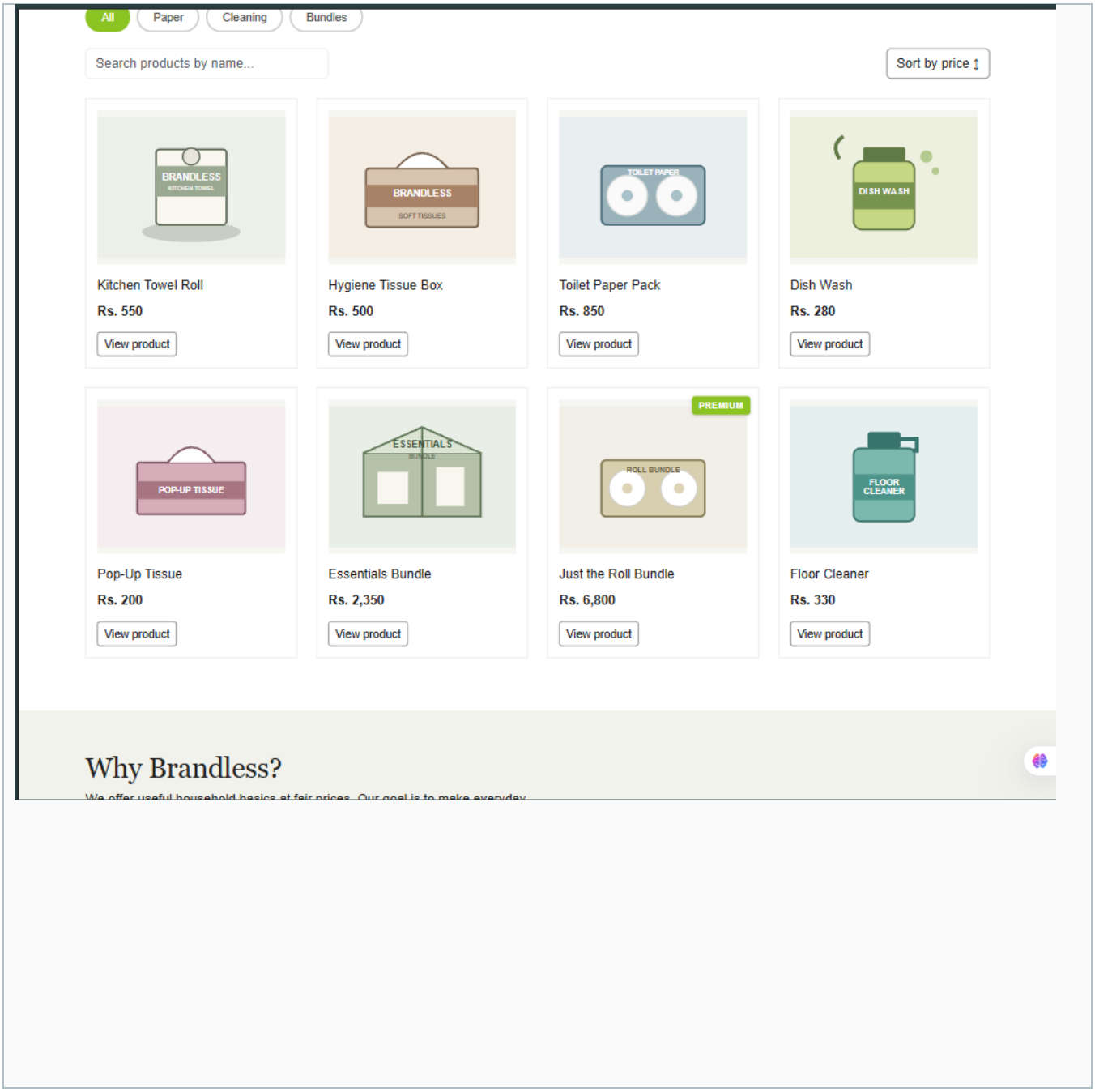
• Task 1 at 663px (BP View)

Listing layout at exactly BP = 663px.



• Task 1 at 1280px (Desktop View)

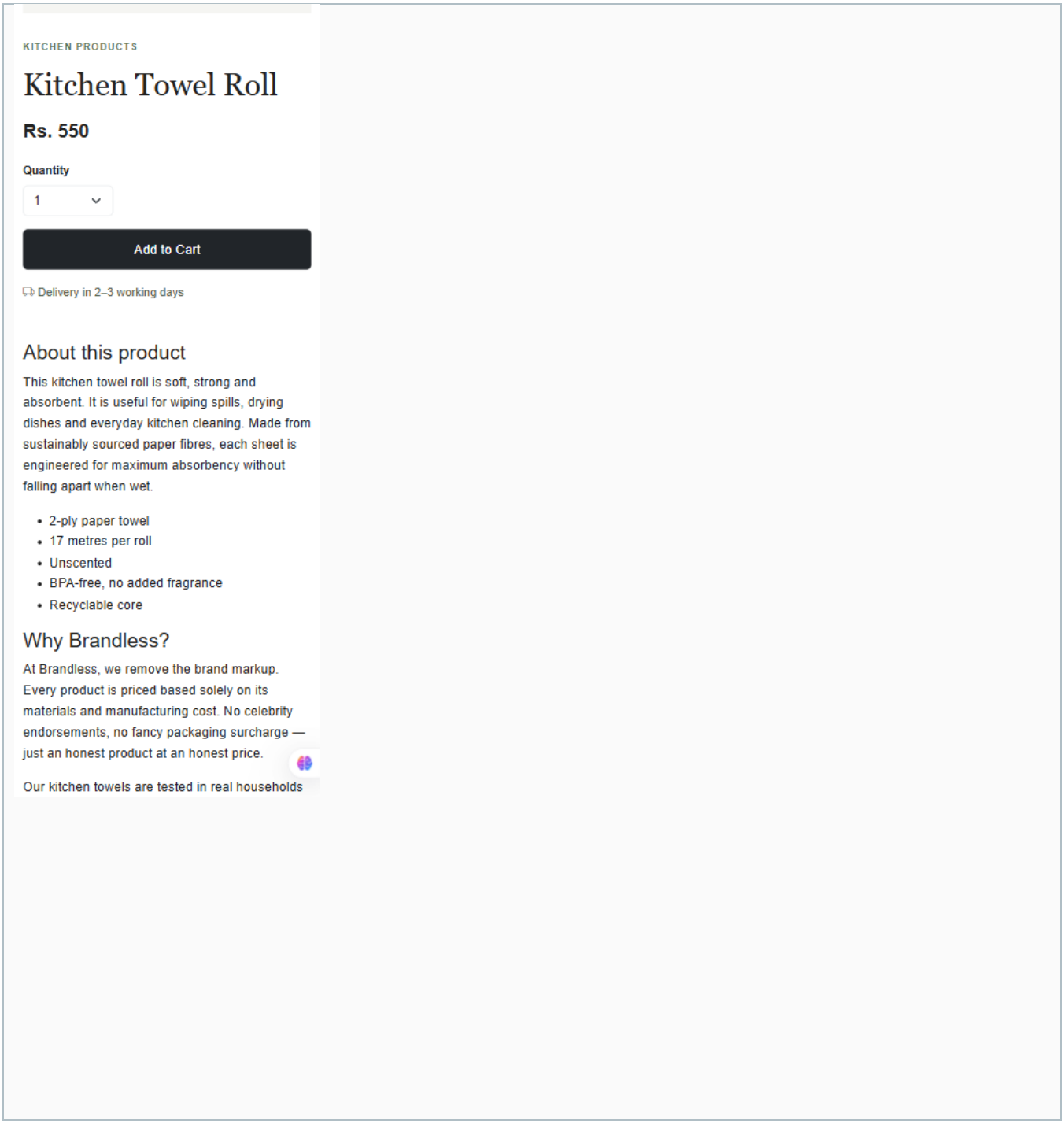
Listing layout on wide screen with all 8 products visible under 'All'.



Task 2 — Product Page Layout with Sticky Details Box Screenshots

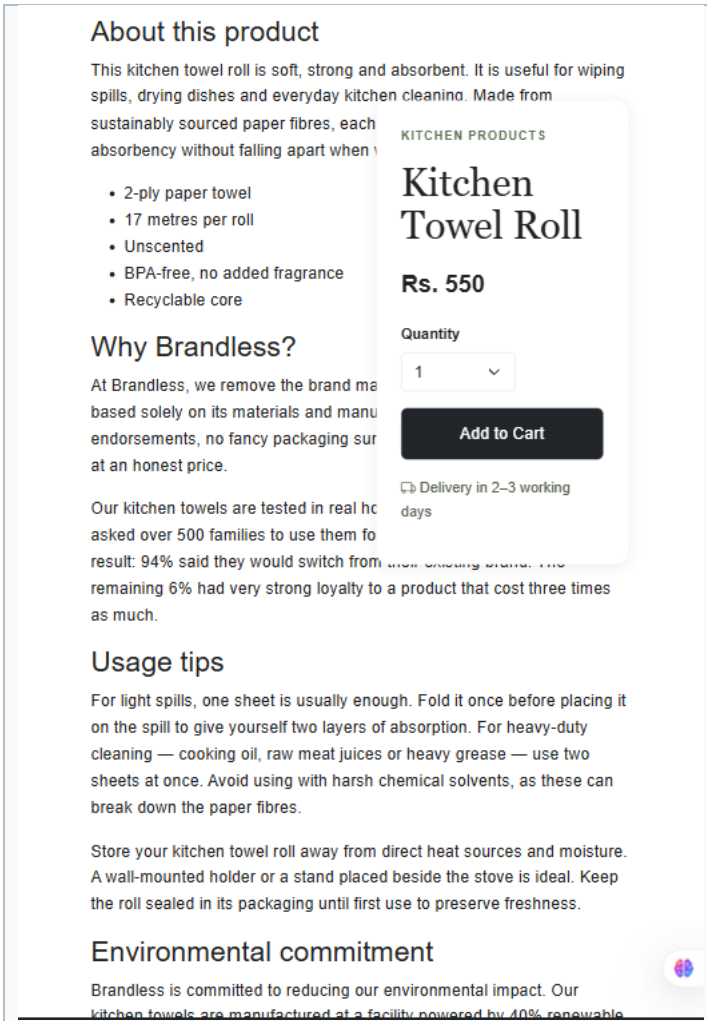
• Task 2 at 375px (Mobile View)

Single-column stacked layout: Image -> Details container -> Description.



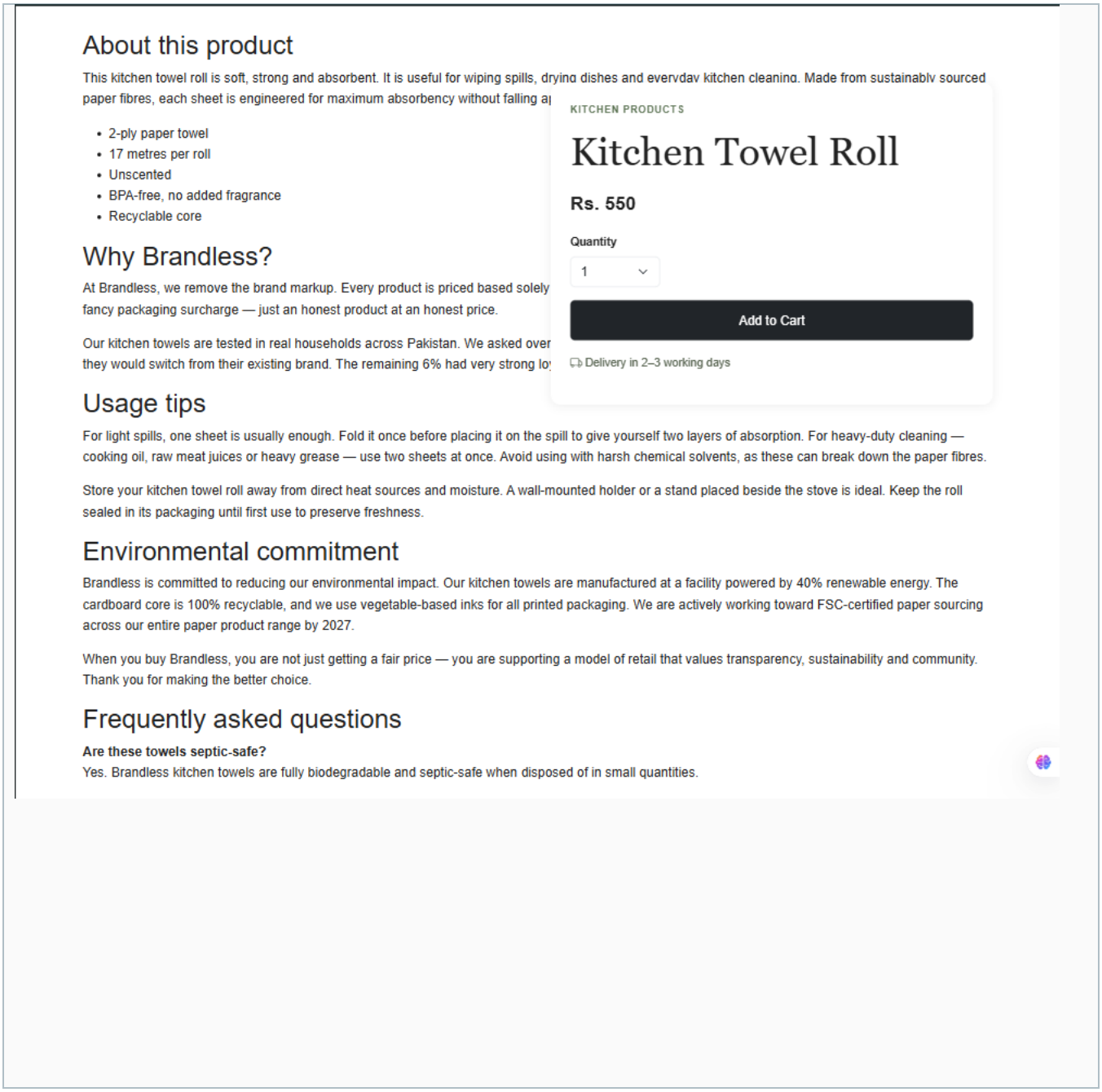
• Task 2 at 663px (BP View)

Two-column grid layout at breakpoint BP = 663px (Image and Details side-by-side, Description full-width below).



• Task 2 at 1280px (Desktop View)

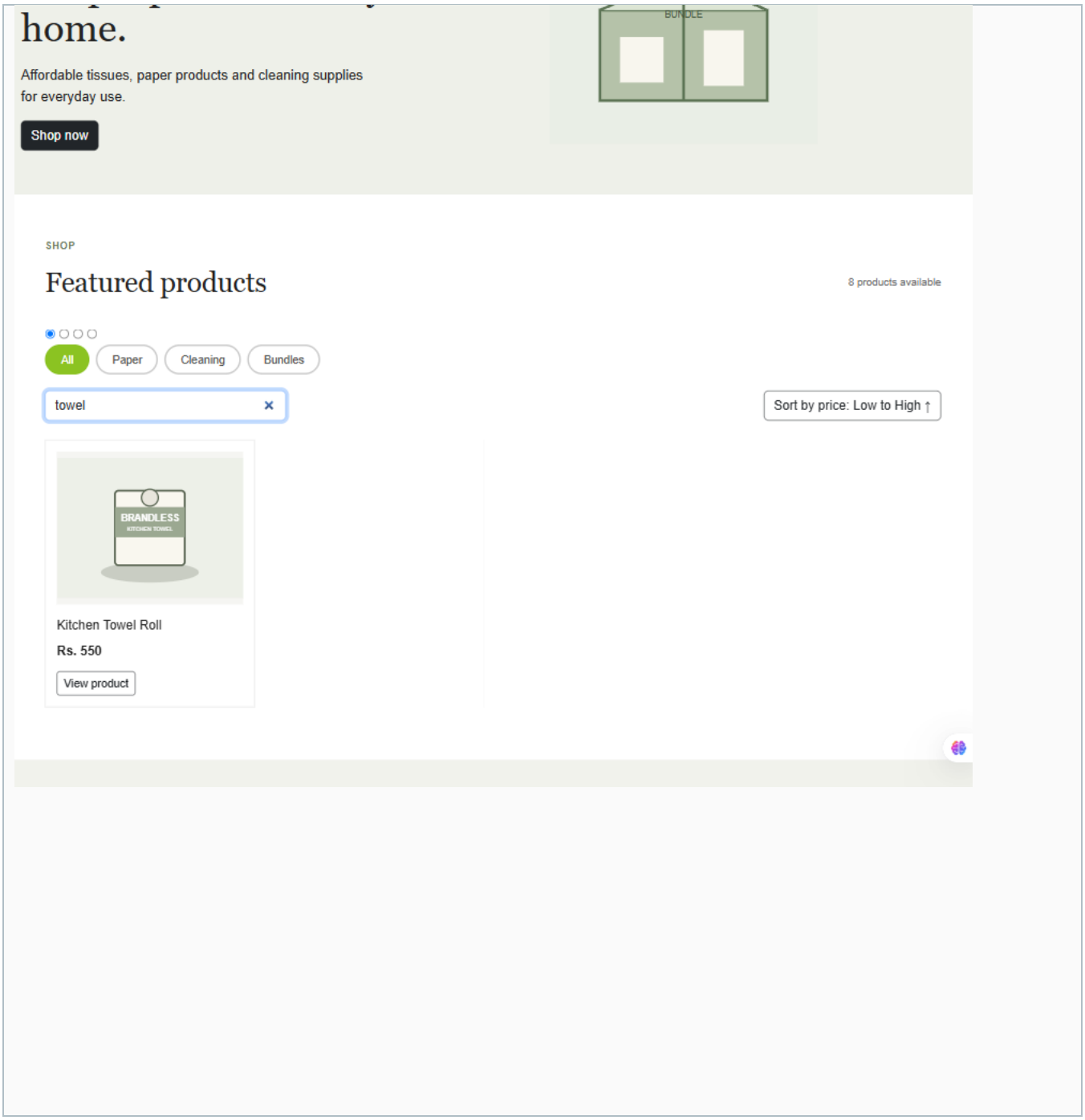
Two-column grid layout on desktop screen.



Task 3 — JavaScript Live Search, Price Sort, and Premium Label Screenshots

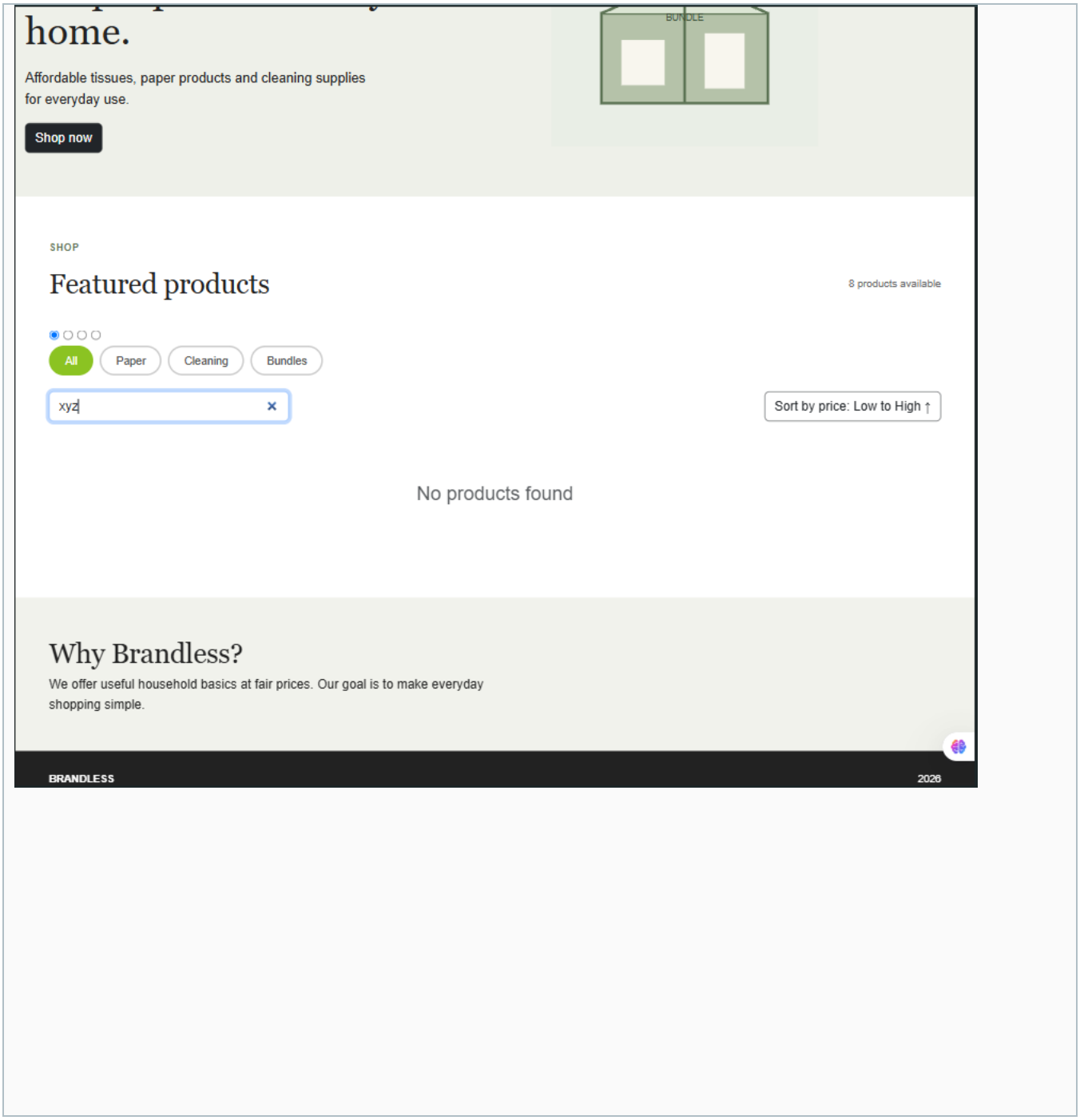
• Task 3 Live Search in Action

Typing in the search box (e.g. 'towel') filtering cards in real time.



• Task 3 'No products found' State

When search query has zero matches (e.g. 'xyz'), showing the message in place of the grid.

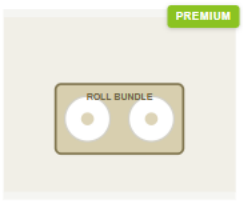


• Task 3 Price Sort in Action

Cards re-ordered from lowest to highest price (or highest to lowest) after clicking 'Sort by price'.

Search products by name...

Sort by price: High to Low ↓



Just the Roll Bundle

Rs. 6,800

View product



Essentials Bundle

Rs. 2,350

View product



Toilet Paper Pack

Rs. 850

View product



Kitchen Towel Roll

Rs. 550

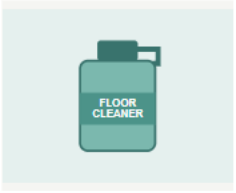
View product



Hygiene Tissue Box

Rs. 500

View product



Floor Cleaner

Rs. 330

View product



Dish Wash

Rs. 280

View product



Pop-Up Tissue

Rs. 200

View product

Why Brandless?

We offer useful household basics at fair prices. Our goal is to make everyday

PART 2: SOURCE CODE ADDED OR CHANGED

Task 1: Markup added to index.html (Category Filter & data-category Attributes)

```
<!-- Radio inputs as direct siblings of #product-grid so ~ sibling selector works -->
<input type="radio" name="cat" id="cat-all"      value="all"      checked>
<input type="radio" name="cat" id="cat-paper"    value="paper">
<input type="radio" name="cat" id="cat-cleaning" value="cleaning">
<input type="radio" name="cat" id="cat-bundles"  value="bundles">

<div class="filter-bar">
  <label for="cat-all"      class="filter-label">All</label>
  <label for="cat-paper"    class="filter-label">Paper</label>
  <label for="cat-cleaning" class="filter-label">Cleaning</label>
  <label for="cat-bundles"  class="filter-label">Bundles</label>
</div>

<!-- Each card wrapped in .col with its respective category attribute -->
<div class="row row-cols-1 row-cols-sm-2 row-cols-lg-4 g-4" id="product-grid">
  <div class="col" data-category="paper"> ... </div>
  <div class="col" data-category="cleaning"> ... </div>
  <div class="col" data-category="bundles"> ... </div>
</div>
```

Task 1: CSS added to styles.css (Pure CSS Category Filter & Active Styling)

```
/* Visually hide radio inputs while keeping them in flow for ~ combinator */
#cat-all, #cat-paper, #cat-cleaning, #cat-bundles {
  position: absolute;
  opacity: 0;
  width: 0;
  height: 0;
  pointer-events: none;
}

/* Category filter label pills */
.filter-bar {
  display: flex;
  flex-wrap: wrap;
  gap: 8px;
  margin-bottom: 20px;
  align-items: center;
}

.filter-label {
  display: inline-block;
  padding: 6px 18px;
  border: 2px solid #ccc;
  border-radius: 999px;
  cursor: pointer;
  font-size: 14px;
  font-weight: 500;
  color: #555;
  transition: border-color .15s, background .15s, color .15s;
  user-select: none;
}
```

```
/* Active filter pill using Accent hue H=81 (hsl(81, 70%, 45%)) */
#cat-all:checked      ~ .filter-bar .filter-label[for="cat-all"],
#cat-paper:checked    ~ .filter-bar .filter-label[for="cat-paper"],
#cat-cleaning:checked ~ .filter-bar .filter-label[for="cat-cleaning"],
#cat-bundles:checked  ~ .filter-bar .filter-label[for="cat-bundles"] {
  background: hsl(81, 70%, 45%);
  border-color: hsl(81, 70%, 45%);
  color: #fff;
}

/* Category filtering rules: hide all, then reveal matching category */
#cat-paper:checked    ~ #product-grid .col { display: none; }
#cat-paper:checked    ~ #product-grid .col[data-category="paper"] { display: block; }

#cat-cleaning:checked ~ #product-grid .col { display: none; }
#cat-cleaning:checked ~ #product-grid .col[data-category="cleaning"] { display: block; }

#cat-bundles:checked  ~ #product-grid .col { display: none; }
#cat-bundles:checked  ~ #product-grid .col[data-category="bundles"] { display: block; }
```

Task 2: Markup in product.html (Named Grid Areas Structure)

```
<main class="container detail-page">
  <a class="back" href="index.html">← Back to products</a>

  <!-- CSS Grid with named areas: image, details, desc -->
  <div class="product-layout">

    <!-- Area: image -->
    <div class="product-image-area">
      <div class="detail-image">
        <img src="assets/kitchen-towel.svg" alt="Kitchen Towel Roll">
      </div>
    </div>

    <!-- Area: details (sticky at BP=663px+) -->
    <aside class="product-details-area">
      <p class="small-title">Kitchen products</p>
      <h1>Kitchen Towel Roll</h1>
      <p class="price">Rs. 550</p>
      <label for="quantity">Quantity</label>
      <select class="form-select" id="quantity">
        <option>1</option>
        <option>2</option>
        <option>3</option>
      </select>
      <button class="btn btn-dark add-button">Add to Cart</button>
      <p class="delivery"><i class="bi bi-truck"></i> Delivery in 2–3 working days</p>
    </aside>

    <!-- Area: desc (spans full width underneath) -->
    <section class="product-desc-area">
      <h2>About this product</h2>
      <p>This kitchen towel roll is soft, strong and absorbent...</p>
      ... <!-- Long multi-section description ensuring page scrollability -->
    </section>
```

```
  </div>
</main>
```

Task 2: CSS added to styles.css (CSS Grid with Named Areas & Sticky Details)

```
/* Product Detail Page - CSS Grid with named grid-template-areas */
.product-layout {
  display: grid;
  grid-template-areas:
    "image"
    "details"
    "desc";
  gap: 32px;
}

.product-image-area   { grid-area: image; }
.product-details-area { grid-area: details; }
.product-desc-area    { grid-area: desc; }

/* Wide screen layout from BP = 663px (600 + N, where N = 63) */
@media(min-width: 663px) {
  .product-layout {
    grid-template-columns: 1fr 1fr;
    grid-template-areas:
      "image   details"
      "desc    desc";
  }

  /* Sticky details box:
     Offset = (63 mod 40) + 70 = 93px from top.
     align-self: start prevents item from stretching to row height.
     Opaque background and rounded corners ensure clean readability over scrolling text. */
  .product-details-area {
    position: sticky;
    top: 93px;
    align-self: start;
    background: #fff;
    border-radius: 12px;
    padding: 24px;
    box-shadow: 0 2px 12px rgba(0, 0, 0, .07);
  }
}

@media(max-width: 662px) {
  .detail-image img {
    height: 280px;
  }
}
```

Task 3: Markup added to index.html (Search/Sort Bar, No Results, Script)

```
<!-- Task 3: Live search and price sort controls -->
<div class="search-sort-bar d-flex flex-wrap gap-2 justify-content-between align-items-center mb-4">
  <input type="search" id="search-box" class="form-control" style="max-width: 300px;"
```

```
         placeholder="Search products by name..." aria-label="Search products">
  <button id="sort-btn" class="btn btn-outline-dark" type="button">Sort by price ↕</button>
</div>

<!-- Task 3: No products found message in place of grid -->
<div id="no-results" class="text-center py-5" style="display: none;">
  <p class="fs-4 text-muted mb-0">No products found</p>
</div>

<!-- Product costing more than N*100 (6,300) gets Premium label -->
<div class="col" data-category="bundles">
  <article class="product-card">
    <img src="assets/roll-bundle.svg" alt="Just the Roll Bundle">
    <h3>Just the Roll Bundle</h3>
    <p>Rs. 6,800</p>
    <a href="product.html" class="btn btn-outline-dark btn-sm">View product</a>
  </article>
</div>

<!-- Script tag linked right before closing body -->
<script src="script.js"></script>
```

Task 3: CSS added to styles.css (Product Card Positioning & Premium Badge)

```
.product-card {
  border: 1px solid #ddd;
  padding: 14px;
  height: 100%;
  background: #fff;
  position: relative;
}

/* Task 3 — Premium Badge (using Accent hue H=81) */
.premium-label {
  position: absolute;
  top: 10px;
  right: 10px;
  background-color: hsl(81, 70%, 45%);
  color: #fff;
  font-size: 11px;
  font-weight: bold;
  text-transform: uppercase;
  padding: 3px 8px;
  border-radius: 4px;
  letter-spacing: 0.5px;
  box-shadow: 0 1px 4px rgba(0, 0, 0, 0.2);
  z-index: 2;
}
```

Task 3: Full JavaScript Implementation (script.js)

```
function initTask3() {
  const productGrid = document.getElementById("product-grid");
  const searchInput = document.getElementById("search-box");
  const sortBtn = document.getElementById("sort-btn");
```

```
const noResults = document.getElementById("no-results");
const categoryRadios = document.querySelectorAll('input[name="cat"]');

if (!productGrid) return;

const productCols = Array.from(productGrid.children);

// 1. Premium label for products costing > N * 100 (63 * 100 = 6,300)
const PREMIUM_THRESHOLD = 63 * 100;

function parsePrice(col) {
  const priceEl = col.querySelector(".product-card p");
  if (!priceEl) return 0;
  // Strip "Rs." and commas before parsing
  const cleaned = priceEl.textContent.replace(/Rs\.\s*/i, "").replace(/,/g, "").trim();
  const num = parseFloat(cleaned);
  return isNaN(num) ? 0 : num;
}

function getProductName(col) {
  const titleEl = col.querySelector(".product-card h3");
  return titleEl ? titleEl.textContent.trim().toLowerCase() : "";
}

// Inject Premium badges on page load
productCols.forEach(function (col) {
  const price = parsePrice(col);
  if (price > PREMIUM_THRESHOLD) {
    const card = col.querySelector(".product-card");
    if (card && !card.querySelector(".premium-label")) {
      const badge = document.createElement("span");
      badge.className = "premium-label";
      badge.textContent = "Premium";
      card.appendChild(badge);
    }
  }
});

// 2. Filter logic (Combines Task 1 Category Filter + Live Search)
function getActiveCategory() {
  const checked = document.querySelector('input[name="cat"]:checked');
  return checked ? checked.value : "all";
}

function filterProducts() {
  const query = searchInput ? searchInput.value.trim().toLowerCase() : "";
  const activeCategory = getActiveCategory();
  let visibleCount = 0;

  productCols.forEach(function (col) {
    const name = getProductName(col);
    const category = col.getAttribute("data-category");

    const matchesSearch = query === "" || name.includes(query);
    const matchesCategory = activeCategory === "all" || category === activeCategory;

    if (matchesSearch && matchesCategory) {
      col.style.display = "";
      visibleCount++;
```

```
      } else {
        col.style.display = "none";
      }
    });

    // If no card matches, show "No products found" in place of the grid
    if (visibleCount === 0) {
      productGrid.style.display = "none";
      if (noResults) noResults.style.display = "block";
    } else {
      productGrid.style.display = "";
      if (noResults) noResults.style.display = "none";
    }
  }

  if (searchInput) {
    searchInput.addEventListener("input", filterProducts);
  }

  categoryRadios.forEach(function (radio) {
    radio.addEventListener("change", filterProducts);
  });

  // 3. Sort by price button (Toggle between ascending and descending)
  let sortAscending = true;

  if (sortBtn) {
    sortBtn.addEventListener("click", function () {
      const currentCards = Array.from(productGrid.children);
      currentCards.sort(function (a, b) {
        const priceA = parsePrice(a);
        const priceB = parsePrice(b);
        return sortAscending ? priceA - priceB : priceB - priceA;
      });

      // Move existing card elements into new order with appendChild (no innerHTML rewrite)
      currentCards.forEach(function (col) {
        productGrid.appendChild(col);
      });

      // Update button label
      sortBtn.textContent = sortAscending
        ? "Sort by price: Low to High ↑"
        : "Sort by price: High to Low ↓";

      // Toggle for next click
      sortAscending = !sortAscending;
    });
  }

  // Initial pass
  filterProducts();
}

if (document.readyState === "loading") {
  document.addEventListener("DOMContentLoaded", initTask3);
} else {
  initTask3();
}
```